



INDAH NOVITA TANGDILILING

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SUMMARY

A determined Informatics student at Institut Teknologi Bandung with a strong passion for using technology to create positive social impact. I enjoy doing teamwork and am committed to continuous personal growth and meaningful contribution. I am currently interested in Cyber Security and DevOps projects and planning on having specialization titles in my next semester.

EDUCATION

Bachelor of Informatics Engineering, Institut Teknologi Bandung

Aug 2023 - Present

- Current GPA of 3.56/4.00

PROJECTS

Developer, Luma_OSjang - Custom OS

- Developed a 32-bit custom operating system in C and Assembly, implementing protected-mode bootstrapping, interrupt handling, keyboard driver support, and a 4 KB kernel stack
- Designed and integrated a block-based EXT2 filesystem and ATA PIO disk I/O layer for a 4 MB storage image using 512-byte blocks

Developer, TCP-Like Transport Layer

- Designed and implemented a custom transport layer protocol in Python over UDP to enable reliable end-to-end communication, including 3-way handshake, sequence/ACK tracking, and orderly connection termination
- Implemented packet segmentation and reassembly with a 512-byte maximum payload, a 17-byte header, and 16-bit checksum validation to detect corrupted packets before delivery
- Added sliding-window retransmission logic with a 4-frame window and timeout-based recovery to maintain data consistency across unstable networks and multi-client client-server communication

Developer, Nimons360 - Family Monitoring App

- Developed a secure Android family-tracking application using Jetpack Compose, implementing 6 core features and 4 bonus features, including real-time GPS tracking via WebSockets, livestreaming with LiveKit, and offline map support with osmdroid
- Built JWT-based authentication and encrypted local storage using EncryptedSharedPreferences and Room to protect sensitive location and family data
- Designed and delivered the application as part of a 3-member team, with modular architecture across authentication, home, map, families, profile, messaging, analytics, and livestream components

EXTRACURRICULAR EXPERIENCE

Deputy Minister of Work Accountability, Kabinet KM ITB

Jul 2025 - Apr 2026

- Oversaw and audited the OKR (Objectives and Key Results) of programs across all ministries within the KM ITB Cabinet

Deputy Head of Welfare, Himpunan Mahasiswa Informatika (HMIF) ITB

Apr 2025 - Mar 2026

- Assisted in managing 30+ staffs and interns focused on supporting student well-being and member services within HMIF ITB
- Identified student needs and provided responsive solutions through welfare-related initiatives and support channels

Speaker, HMIF Leadership Class 2025

Sept 2025

- Invited Speaker for the HMIF Leadership Class, presenting to an audience of 100+ on the intersection of leadership and event operational
- Facilitated an interactive session that translated complex management theories into practical steps for planning and executing large-scale events

Chairman, Sekolah Perangkat Mentor OSKM ITB 2025

May 2025 - Aug 2025

- Led the leadership development program for over 70 OSKM mentors
- Designed and oversaw training curricula focused on leadership, facilitation, project management, and student engagement

Chairman, Wisuda April Himpunan Mahasiswa Informatika (HMIF) ITB

Feb 2025 - Apr 2025

- Led a committee of 100+ members to plan and execute the April graduation ceremony for HMIF ITB
- Set project direction, managed timelines, and facilitated inter-division communication to maintain clarity and alignment throughout the preparation process
- Supervised budgeting, resource allocation, and stakeholder coordination to ensure an efficient and well-organized event

Head of Mentors, Kaderisasi Wilayah STEI-K ITB 2024

Dec 2024 - Jan 2025

- Led the coordination, training, and performance management of 50+ mentors responsible for guiding 300+ new students through different majors in the faculty
- Collaborated with various committees to ensure a supportive, structured, and impactful experience.

SKILLS

• Hard Skills

- **Programming Languages:** Python, Java, C/C++, TypeScript, JavaScript, Go (Golang), Kotlin, SQL, Bash/Shell, Assembly
- **Web & Mobile Development:** Next.js, React.js, Node.js, TailwindCSS, Flask/Django, Android SDK, Jetpack Compose
- **Data Science & AI/ML:** NumPy, Pandas, Scikit-Learn, Matplotlib, Neural Networks (FFNN), SVM, Predictive Modeling, Optimization Algorithms (GA, Simulated Annealing)
- **Databases & Infrastructure:** PostgreSQL, MySQL, Docker, Room (Android)
- **Tools & Technologies:** Git/GitHub, REST APIs, WebSockets, JWT Authentication, LiveKit, Maven, Jupyter Notebook
- **Core Competencies:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Multithreading, OS/Memory Management, System Architecture (MVC/MVVM, Plugin)

- **Soft Skills:** Public Speaking, Effective Communication, Event & Project Management, Leadership & Team Management, Strategic Planning, Problem Solving